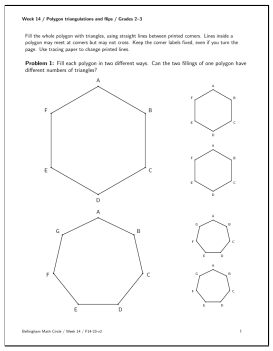
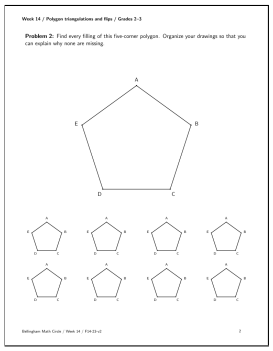
# Week 14 / Polygon triangulations and flips / Problem menu

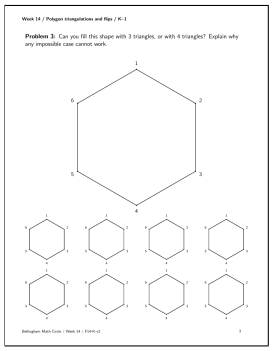
for adults: pick, then print



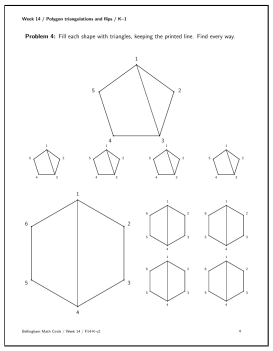

**Two fillings of each polygon**
K–1 P1 · Grades 2–3 P1
first route
*wording differs by band*



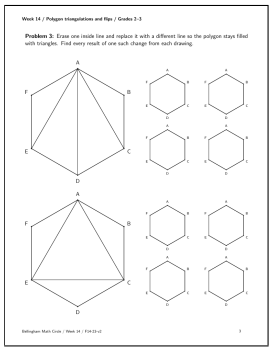

**Every filling of the pentagon**
K–1 P2 · Grades 2–3 P2 · Grades 4–5 P1
first route
*wording differs by band*



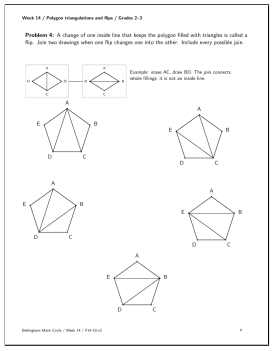

**Three triangles or four in the hexagon**
K–1 P3
first route



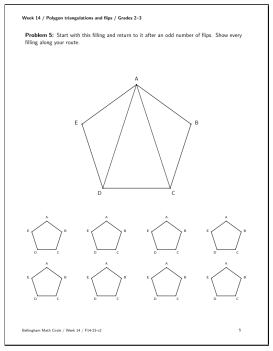

**Completions that keep a printed line**
K–1 P4
return route



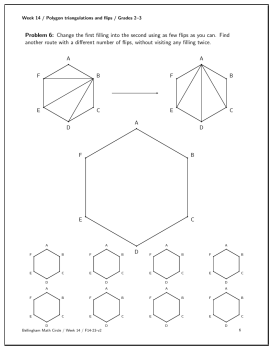

**Every result of one change**
K–1 P5 · Grades 2–3 P3 · Grades 4–5 P2
K-1 return; 2-3 first; 4-5 first
route
*wording differs by band*



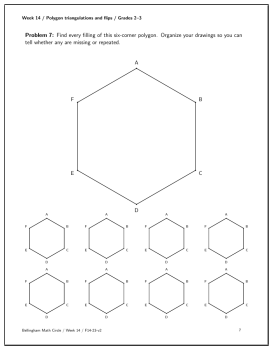

**The flip map of the pentagon**
Grades 2–3 P4 · Grades 4–5 P3
2-3 return; 4-5 first route
*wording differs by band*



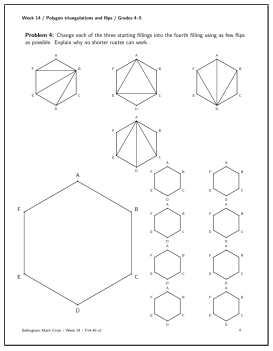

**A loop through the pentagon fillings**
K–1 P6 · Grades 2–3 P5
return route
*wording differs by band*



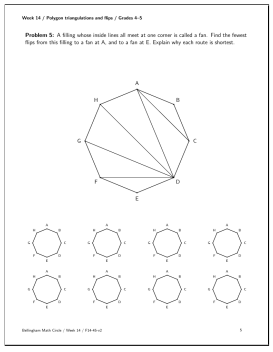

**Fewest flips between two hexagon fans**
Grades 2–3 P6
return route



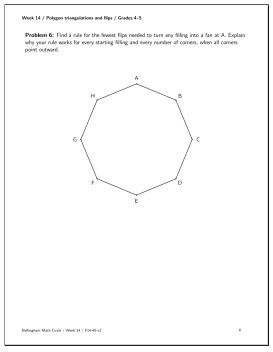

**Every filling of the hexagon**
Grades 2–3 P7
return route



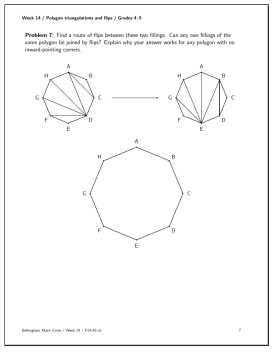

**Three starts to one hexagon fan**
Grades 4–5 P4
return route




**Fewest flips to two octagon fans**
Grades 4–5 P5
return route




**A rule for the fewest flips to a fan**
Grades 4–5 P6
return route




**Any two fillings are joined by flips**
Grades 4–5 P7
return route